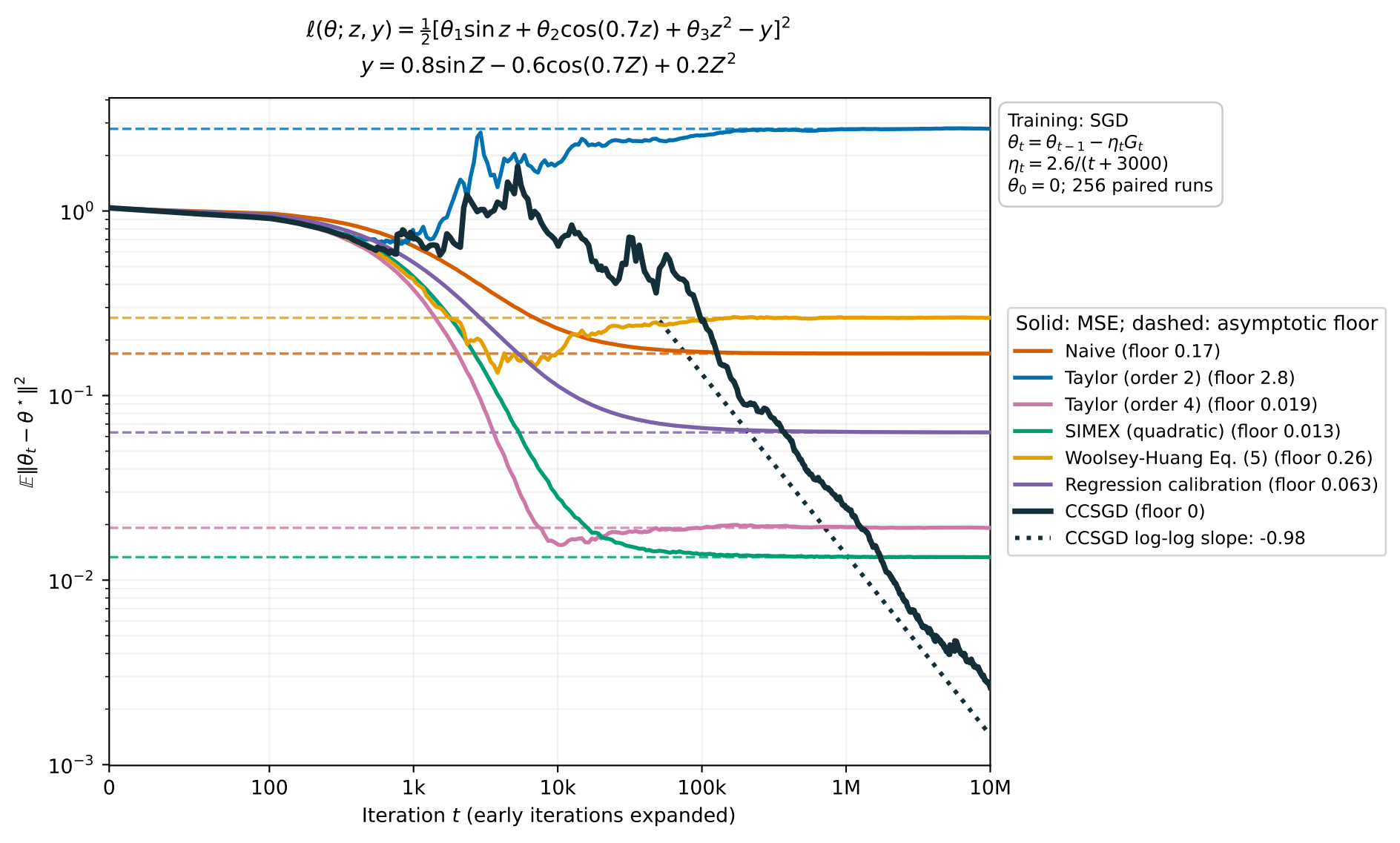
$\ell(\theta; z, y) = \frac{1}{2}[\theta_1 \sin z + \theta_2 \cos(0.7z) + \theta_3 z^2 - y]^2$
$y = 0.8 \sin Z - 0.6 \cos(0.7Z) + 0.2Z^2$
$\mathbb{E}\|\theta_t - \theta^\star\|^2$
10^0
10^{-1}
10^{-2}
10^{-3}
0
100
1k
10k
100k
1M
10M
Iteration t (early iterations expanded)
Training: SGD
$\theta_t = \theta_{t-1} - \eta_t G_t$
$\eta_t = 2.6/(t + 3000)$
$\theta_0 = 0$; 256 paired runs
Solid: MSE; dashed: asymptotic floor
Naive (floor 0.17)
Taylor (order 2) (floor 2.8)
Taylor (order 4) (floor 0.019)
SIMEX (quadratic) (floor 0.013)
Woolsey-Huang Eq. (5) (floor 0.26)
Regression calibration (floor 0.063)
CCSGD (floor 0)
CCSGD log-log slope: -0.98